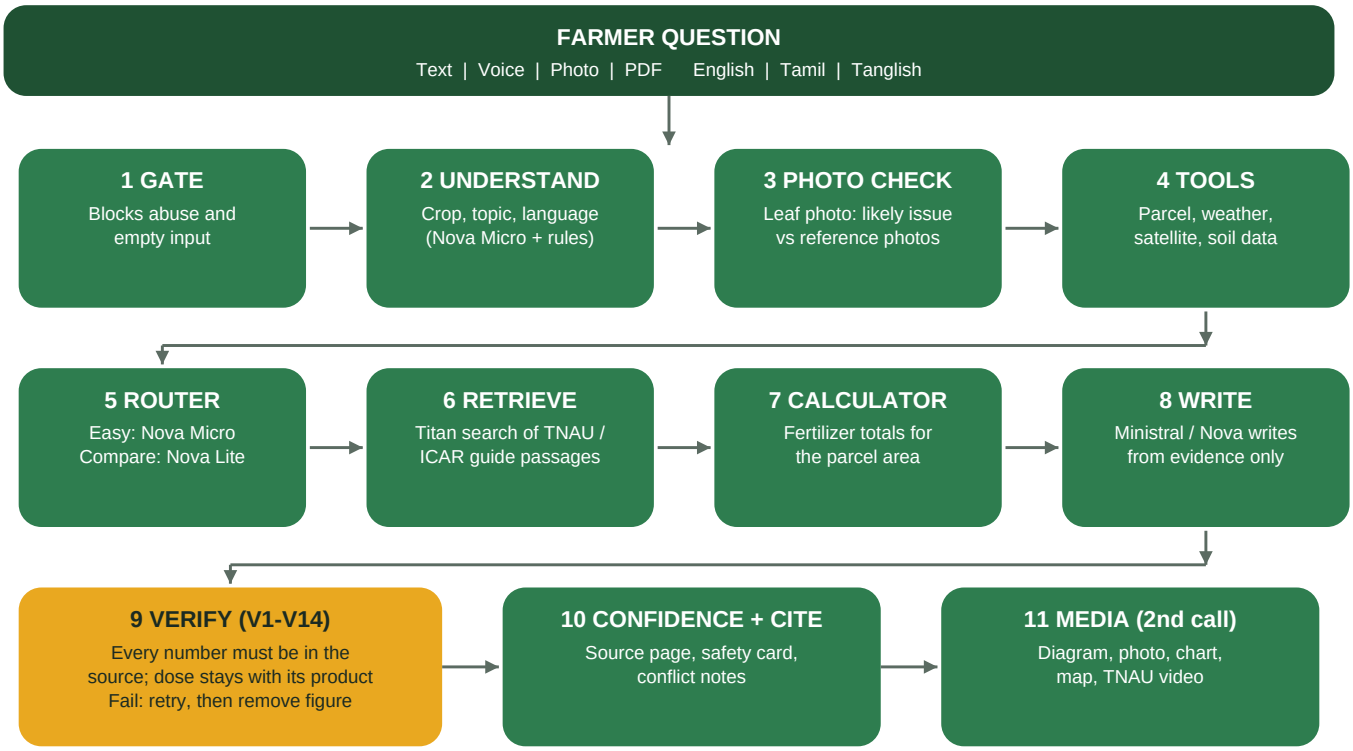
Cropsy

Judge briefing: pipeline, shadow accuracy, 16 map layers, token proof and demo questions

Team 36 | English, Tamil and Tanglish | Text, photo, PDF

1. How Cropsy answers: pipeline schematic

Every question goes through the same eleven steps. Green boxes do a job; the gold box is the safety check that makes Cropsy different from a normal chatbot.



Text answer is shown first. Media loads after, so the farmer never waits for pictures.

Models (AWS Bedrock, ap-south-1): Nova Micro / Lite, Ministral 3B / 8B, Titan embeddings. Textract reads PDFs.

2. Why we built the pipeline this way

Step	What it does	Why we use it
Gate + understand	Cleans the question, finds the crop, topic, parcel and language (rules plus Nova Micro).	Farmers write in Tamil, Tanglish and short phrases. Rules fix crop and topic words the model may miss, so cheap models stay reliable.
Photo check	Compares a leaf photo with reference photos and states a possible, not confirmed, issue.	A photo gives symptoms the farmer cannot describe. We never present it as a diagnosis.
Tools	Reads parcel soil, weather, satellite indices, sunlight and terrain.	Real data about the farmer's own land, not a decorative image.
Router	Easy questions go to Nova Micro; comparisons go to Nova Lite.	Keeps cost low. The measured cost of one text answer was about $0.000076.
Retrieve	Titan embeddings search TNAU and ICAR passages by meaning.	The answer comes from trusted guides, so it can be traced to a page.
Calculator	Parcel fertilizer totals: per-hectare dose x area (urea 46% N, DAP 18-46-0, MOP 60% K2O).	A language model must not do arithmetic that affects money and crops. The calculator is exact and tested.
Write	Ministral / Nova writes only from the retrieved passages.	Limits invention. If no evidence exists, a safe general answer is written instead and labelled as such.
Verify V1-V14	Checks every figure: in the source, right unit and basis, right purpose, dose tied to its own product.	Language models can state a wrong number confidently. Wrong pesticide doses are the highest-risk error, so we catch them before the farmer sees them.
Cite + safety	Shows the source page; banned or restricted chemicals get a fixed safety card with no dose.	Trust and safety. The farmer can check the page, and is sent to the KVK for restricted chemicals.
Media (2nd call)	Adds a diagram, a licensed photo, a chart or a verified video if one fits.	Pictures explain faster than text. It is a separate call so the text appears first.

Current results: 267 automated tests pass. Our 20-question live test set scored 20/20 on numbers, citations and behaviour in 4 of 4 runs. Expected answers were read from the source by the team, not by an agronomist; expert review is the next step.

3. Shadow accuracy: measuring real sunlight on each parcel

The map shows where hills block the sun on each parcel, hour by hour. The land is in a hill-foot village (Then Malai), so a ridge to the west can cut the afternoon sun by hours. Crops need to be chosen and planned around that.

Measurement	Advantage for the farmer
Mountain shadows	Finds areas with less direct sunlight, so shade-tolerant crops go there.
Cloud cover	Estimates temporary reductions in sunlight (shown as a cloud forecast layer from a ~10 km weather model).
Sunlight intensity and duration	Improves crop-growth estimates (direct-sun hours for four dates: 21 Mar, 21 Jun, 23 Sep, 21 Dec).
Combined terrain and weather data	Supports better crop selection and farm planning.

How we computed it

A solar-position calculation (NOAA algorithm) gives the sun's height and direction for any minute. A 30 m Copernicus elevation model gives each parcel a horizon profile: the highest hill angle in every compass direction. Sun below the horizon angle means the parcel is in shadow. Direct-sun hours are the minutes the sun is above that profile.

Proof: independent checks (all passing, 19 automated tests)

What we compared	Against	Result required (test passes)
Sun height and compass direction	suncalc, an independent solar library, on 6 dates and times	Height within 0.3 degrees, direction within 0.6 degrees
Sunrise and sunset	suncalc	Within 3 minutes
Atmospheric bending of light	Physics reference	About 0.5 degrees at the horizon, almost none overhead
Shadow length on a test ridge	Geometry: height / tan(sun altitude)	Within 10%
Highest hill angle at each anchor parcel	PVGIS (European Commission) horizon data	Within 3 degrees
Ground height at each parcel	Stored elevation values	Within 5 m
Shadow map vs parcel horizon profiles	Each other (consistency)	Agree at the parcels
Flat ground	Should have no shadow	No shadow

Honest limits: the terrain data resolution is 30 m (90 m for the wide view), so a small tree or building is not modelled. Cloud is a forecast from a coarse (~10 km) model, not a measurement at the field. The sunlight-hours table is terrain only; cloud is a separate toggle. We have not yet compared the numbers with a light meter in a real field.

4. Why 16 map layers help (and how we know they are not wrong)

One image cannot tell a farmer why a field looks weak. Sixteen layers look at the same field in different ways: how green, how wet, how much water the leaf holds, what the soil is, what the ground is, and what radar sees through clouds. When two layers agree, the signal is stronger; when they disagree, it is a reason to walk the field. Each layer is labelled real, modelled or derived.

Group	Layers	What the farmer learns	Kind
Crop health	NDVI, EVI, SAVI, NDRE	How green and healthy the crop is. EVI and NDRE do better in dense canopy, SAVI in young crops.	Real (Sentinel-2)
Water	NDWI, NDMI, ET	Open or very wet ground, water inside the leaf, and estimated daily crop water use.	NDWI, NDMI real; ET modelled
Soil	SOC	Soil organic carbon: a proxy for fertility (250 m, indicative).	Real (SoilGrids model)
Terrain and land	DEM, LULC	Ground height (drainage, hill shade) and what covers the ground (check a parcel is really cropland).	Real
Display	TCI, ETCI, Hybrid, Colour-blind NDVI	The village as seen from space, with greenness overlaid; a palette readable with red-green colour blindness.	TCI real; others derived
Radar	RVI, RSM	Crop structure and soil wetness even under cloud (Sentinel-1).	Real; RSM is relative

Proof checks

Check	Result
Sanity against ESA WorldCover land cover: open water	NDVI about 0, NDWI +0.08 (water is wettest, least green)
Tree cover	Highest NDVI, 0.63
Cropland	NDVI 0.40
Built-up	NDVI 0.31
Build script assertion	Every index mean must lie between -1 and 1
Known fault caught and fixed	A scene with the wrong reflectance offset gave NDVI 0.9+. Offset is now detected from the pixels, not trusted from metadata

Honest limits: a falling index means go and look, not a diagnosis. Crop names, stages and issues in the demo village are dummy data; the satellite sees what really grows at the coordinates. Radar soil moisture is noisy under dense canopy.

5. Photos and tokens: what does a picture cost?

Question: when Cropsy answers and also shows a photo, does the photo use AI tokens? We measured it.

What	Measured result (run on this machine)
Choosing a visual for 'how to prepare vermicompost'	1 Nova Micro call, 708 tokens. Result: vermicompost photo plus compost-bed diagram.
Choosing a visual for 'wheat seed rate per hectare'	1 Nova Micro call, 705 tokens. Result: no visual (none fits).
Choosing a visual for 'rice blast symptoms'	1 Nova Micro call, 703 tokens. Result: photo plus table.
The photo file itself	0 model tokens. It is a licensed image stored with the app and sent as a file.
Diagrams, charts, satellite panel, weather chart	0 model tokens. Drawn by program code from real data.
The text answer (wheat seed rate)	Estimated cost $0.000076 reported by the app (Nova Micro).

So a picture costs about 700 tokens once, for deciding which picture fits, not for the picture. That decision runs after the text is shown, so it never slows the answer. If that call fails, simple keyword rules choose the visual at no model cost. When nothing fits, no visual is shown and no wrong picture is forced in.

Not measured: when a farmer uploads their own leaf photo, an image model reads it. That costs more than a text question; we have not measured its tokens, so we make no claim about it here.

6. Demo questions that work well

Questions 1-6, 8 and 11 are in our live test sets and passed in the latest runs. The rest were tested in earlier browser sessions. Run **aws sso login** first.

#	Ask this	What it shows
1	wheat seed rate per hectare	A cited answer: 100 kg per hectare, with the guide page.
2	கம்பு விதை அளவு எவ்வளவு	Tamil answer; the nursery figure is never given per hectare.
3	uram evlo podanum nel ku	Tanglish understood as a rice fertilizer question.
4	What is the seed rate for cowpea?	Honest answer: the guides we hold do not give it. No invented number, one follow-up question.
5	ragi blast spray dose	Dose stays with its own product (carbendazim 500 g).
6	ulundhu pooch marundhu	Blackgram pest control, product with its own dose.
7	how much urea and dap for TM-31	Calculator: per-hectare dose and parcel totals shown separately.
8	TM-05 satellite health this week	Real satellite NDVI for the parcel, latest scene date.
9	how to prepare vermicompost	Steps plus photo, diagram and a TNAU video.
10	how does photosynthesis work	General science answered, with a diagram.
11	what is the price of tomato today	Says it does not hold live prices: no invented figure.
12	wheat sowing time, with region set to Punjab	Does not present Tamil Nadu dates as Punjab advice.
13	Upload a leaf photo	A possible issue with reference photos, not a confirmed diagnosis.
14	Map view: Sunlight hours panel, Terrain shadow and Cloud toggles	The shadow feature. This is in the map, not the chat box.

Please skip the voice demo: browser microphone input is not verified. Production decision: hold until the agronomist and Tamil-language reviews are complete.